

1. (a) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g, a.e. (\mu), \forall n \geq 1, \int g d\mu < \infty$ and $f_n \rightarrow f$ a.e. (μ) , then $f \in L^1(\Omega, \mathcal{F}, \mu), \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.
 - (b) We will use this real analysis fact: A sequence $\{x_n\}$ converges to x iff for any subsequence $\{x_{n_k}\}$ of $\{x_n\}$, there is a further subsequence $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ such that $x_{n_{k_l}} \rightarrow x$. Denote $a_n = \int f_n d\mu, a = \int f d\mu$ and $b_n = \int |f_n - f| d\mu$. Will show $a_n \rightarrow a$ and $b_n \rightarrow 0$. Choose any subsequence $\{a_{n_k}\}$ of $\{a_n\}$, and for this subsequence, choose the corresponding subsequence functions $\{f_{n_k}\}$ of $\{f_n\}$. Since $f_n \rightarrow f$ in measure, then $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0, \forall \varepsilon > 0$, thus $\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \varepsilon) = 0, \forall \varepsilon > 0$, i.e. $f_{n_k} \rightarrow f$ in measure. By Thm 2.5.2, we can get a sub-subsequence $f_{n_{k_l}} \rightarrow f$ a.e. (μ) . For this sub-subsequence $\{f_{n_{k_l}}\}$, conditions of DCT are true. Hence from DCT, we have $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu$, i.e. $a_{n_{k_l}} \rightarrow a$. By Fact 1, we get $a_n \rightarrow a$, i.e. $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$. Similarly, we have $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.
 - (c) $f = 1_A$, where A is a non-Borel measurable set.
 - (d) $g = 1_A - 1_{A^C} = \begin{cases} 1; & \text{on } A \\ -1; & \text{on } A^C \end{cases}$ is not measurable, but $|f| \equiv 1$ is Borel measurable
2. (a) Choose $(\Omega, \mathcal{F}, P) = (\mathbb{R}, B(\mathbb{R}), \text{Normal}(0,1))$, i.e. $P(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dm(x)$ and choose $\epsilon(\omega) = \omega$. This (canonical construction) shows ϵ is a $\text{Normal}(0,1)$ random variable.
 - (b) Note that $\{\epsilon_i : i \geq 1\}$ are measurable, and for each $i \geq 1, Y_i$ is created by adding a constant to ϵ_i and hence Y_i is measurable. For each $n \geq 1, \hat{\beta}_n$ is created by multiplication, component-wise addition and division (multiplication) by a constant of measurable functions $\{Y_i : i \geq 1\}$ and hence they are measurable too.
 - (c) Using linearity of Lebesgue integrations, we have $\int Y_i dP = x_i \beta + \int \epsilon dP = x_i \beta$. Also,

$$\int \hat{\beta}_n dP = \frac{1}{\sum_{i=1}^n x_i^2} \int \sum_{i=1}^n x_i Y_i dP = \frac{1}{\sum_{i=1}^n x_i^2} \sum_{i=1}^n x_i \int Y_i dP = \frac{1}{\sum_{i=1}^n x_i^2} \sum_{i=1}^n x_i^2 \beta = \beta$$

3. (i) $\forall \omega \in [0, 1), X_n(\omega) = 0$ for large n such that $\omega < 1 - \frac{1}{n}$. But $X_n(1) = 1 \neq X(0), \forall n \geq 1$. So $X_n(\omega) \rightarrow X(\omega), \forall \omega \in [0, 1)$, and $P(\{1\}) = 0$ So. $X_n \rightarrow X$ a.e. (P) .
- (ii) $m(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = m([1 - \frac{1}{n}, 1]) = \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty, \forall \epsilon > 0$. So $X_n \xrightarrow{p} X$.
- (iii) $(E|X_n - X|^p) = (\int |X_n - X|^p dP) = n^{2p} \cdot m([1 - \frac{1}{n}, 1]) = n^{2p-1} \rightarrow 0$ as $n \rightarrow \infty$ iff $0 < p < \frac{1}{2}$. Hence $X_n \rightarrow X$ in L^p for $0 < p < \frac{1}{2}$, but this doesn't hold for $\frac{1}{2} \leq p < \infty$.
- (iv) Let $g(x) = x^{\frac{p}{q}}, x > 0$ and $\frac{p}{q} > 1$, then $g(x)$ is a convex function. Since $X_n \rightarrow X$ in L^p , then $(\int |X_n - X|^p dP) \rightarrow 0$ as $n \rightarrow \infty$. By Jensen's inequality, we have $g(\int |X_n - X|^q dP) \leq \int g(|X_n - X|^q) dP$, i.e. $(\int |X_n - X|^q dP)^{\frac{p}{q}} \leq \int |X_n - X|^p dP$

$$\text{Then } \int |X_n - X|^q dP \leq \left(\int |X_n - X|^p dP \right)^{\frac{q}{p}} \rightarrow 0, \text{ as } n \rightarrow \infty.$$

i.e. $(\int |X_n - X|^q dP)^{\frac{1}{q} \wedge 1} \rightarrow 0$ as $n \rightarrow \infty$. Hence $X_n \rightarrow X$ in L^q for any $0 < q < p$.